thereby develop his laws of general relativity. And only after Hawking had *defined* the concept of a dynamical hole's horizon could he and others explore in detail how black holes change when pummeled by collisions or by infalling debris.

Before November 1970, most physicists, following Roger Penrose's lead, had thought of a hole's horizon as "the outermost location where photons trying to escape the hole get pulled inward by gravity." This old definition of the horizon was an intellectual blind alley, Hawking realized in the ensuing months, and to brand it as such he gave it a new, slightly contemptuous name, a name that would stick. He called it the *apparent horizon.*[2]

Hawking's contempt had several roots. First, the apparent horizon is a relative concept, not an absolute one. Its location depends on the observers' reference frame; observers falling into the hole might see it at a different location from observers at rest outside the hole. Second, when matter falls into the hole, the apparent horizon can jump suddenly, without warning, from one location to another—a rather bizarre behavior, one not conducive to easy insights. Third and most important, the apparent horizon had no connection at all to the flash of congealing mental pictures and diagrams that had produced Hawking's New Idea.

Hawking's new definition of the horizon, by contrast, was absolute (the same in all reference frames), not relative, so he called it the *absolute horizon.* This absolute horizon is beautiful, Hawking thought. It has a beautiful definition: It is "the boundary in spacetime between events (outside the horizon) that can send signals to the distant Universe and those (inside the horizon) that cannot." And it has a beautiful evolution: When a hole eats matter or collides with another hole or does anything at all, its absolute horizon changes shape and size in a smooth, continuous way, instead of a sudden, jumping way (Box 12.1). Most important, the absolute horizon meshed perfectly with Hawking's New Idea:

Box 12.1

Absolute and Apparent Horizons for a Newborn Black Hole